Mars Chen

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Languages: Mandarin (Native) / English (Professional working proficiency)

PROFESSIONAL SUMMARY

Java backend engineer with 3+ years of commercial experience across three data-sensitive industries — telecom, healthcare, and now finance — building production systems on Java / Spring Boot microservices and the data layer beneath them, with enough front-end reach (Angular) to deliver a whole system alone. Two consistent threads: legacy modernisation in banking (Spring Boot 4 upgrades, migration off IBM WebSphere, cloud migration) and regulatory delivery (PDPA access controls, OTP hardening, audit response). Previously three years as a product manager, working directly with engineering on specification and acceptance.

TECHNICAL SKILLS

- **Backend:** Java (8 / 11 / 17), Spring Boot, Spring Security, Spring Data JPA, Hibernate, Spring Session, jOOQ, RESTful API design, Java EE / JSF, PHP / Laravel, Python, JUnit / PHPUnit / pytest
- **Frontend:** Angular, TypeScript, JavaScript, SCSS / CSS3, jQuery
- **Database:** Oracle, MySQL / MariaDB, SQL Server, ClickHouse, Redis, Memcached
- **DevOps & Platform:** Docker, Kubernetes (Deployment / Ingress), GitHub Actions, Maven, Gradle, Git / Bitbucket, Linux / CLI, Tomcat, Nginx / Ingress

PROFESSIONAL EXPERIENCE

Software Engineer
2025.09 – Present

Twinfo Co., Ltd., Taipei, Taiwan

Software consultancy delivering production systems for enterprise financial clients, primarily Taipei Fubon Bank; work spanned nine banking systems.

- **Architecture & Platform Migration:** Leading the merchant portal's migration off IBM WebSphere 9, which cannot run Spring Boot 3 or 4, onto a two-node embedded Tomcat 11 stack with Java 8 → 17 and Spring Boot 2.7 → 4.1 (go-live expected end of 2026); the PoC root-caused **18 upgrade issues** — **eight** of them compiling and starting cleanly yet silently changing behaviour — and diffed **278 endpoints** before and after.
- **Cloud Migration & Kubernetes:** Leading the municipal-treasury system's migration from on-premises WebSphere to the bank's container platform, upgrading Spring Boot 1.5 / 2.7 to 3+ (go-live expected end of 2026); bytecode analysis of **15,242 classes** in closed-source components narrowed the Jakarta risk to **two** unused code paths. Also implemented the promotions site's Spring Boot 4.1.1 upgrade on multi-pod Kubernetes, resolving **three** issues a single pod cannot surface.
- **Banking System Ownership:** Lead development and maintenance of the back office for the bank's trust signing system, implementing review and approval workflows, deed diff comparison, ESB SOAP integration, field-level encryption, and LDAP authentication single-handedly.
- **Compliance & Security Delivery:** Implemented the PDPA blocked-list check end to end, from the ingest batch through to guard points in **five application flows**, across two codebases that share no code — Spring Boot microservices and Java EE; also moved OTP to RFC 6238 TOTP with a DB-managed key.
- **Client Interface & Scoping:** Act as technical point of contact for security remediation — verify penetration-test and vulnerability-scan findings line by line against the source to confirm they actually apply, discard misreported and overlapping items, then propose the fix and its blast radius for the client to decide on; architectural changes are likewise put to the client with risks and timelines before implementation starts.

Java Backend Engineer
2024.08 – 2025.06

Stork Fertility Center, Taipei, Taiwan

One of Taiwan's largest reproductive medicine centres.

- **System Integration & Architecture:** Integrated three heterogeneous systems — clinical examination, document management, and customer management — into the centre's Artemis platform; refactored data classification logic onto the Strategy Pattern so new categories became additive rather than invasive.
- **Data Layer:** Designed MySQL schemas and optimised query performance via Spring Data JPA and Redis caching.

Java Engineer
2023.03 – 2024.06

Groundhog Technologies Co., Taipei, Taiwan

Telecom network-intelligence software vendor serving mobile carriers worldwide.

- **Performance:** Introduced an R-Tree spatial index in place of brute-force distance scanning in the RF recommendation module, cutting execution time by **over 50%** in benchmarks against the project's test dataset.
- **Core Development:** Built and maintained RF recommendation modules for a mobile network geolocation platform (Java 8, Gradle, Linux); designed the MySQL output schema and unique indexes for large-scale network data, and wired the bulk-load path into ClickHouse for high-volume result sets.

APP Product Manager
2022.05 – 2022.08

D-Link Corporation, Taipei, Taiwan

- Coordinated app product development and launch across the product centre and multiple departments.

Product Manager
2019.10 – 2022.05

Adam Elements International, Taipei, Taiwan

Consumer electronics accessories brand selling into global retail channels.

- **Ownership & Scale:** Owned a core product line contributing **~40% of company revenue**, covering technology research, requirements analysis, and competitive studies.
- **Launch Execution:** Led **8 product launches in 2021** delivering **35% YoY revenue growth**; ran specification and acceptance across engineering, design, supply chain, and production planning.

PROJECTS

Podcast Note Automation Pipeline
2026

Personal end-to-end automation: one LINE message produces a full episode digest, unattended.

- **Pipeline:** LINE bot → Google Apps Script trigger queue → polling agent on macOS → RSS download → local Whisper transcription → LLM proofread and summary → HTML cards delivered back to LINE. **~50 episodes processed**.
- **Reliability & Tooling:** Lock-file mutual exclusion, heartbeat write-back, and sleep-aware scheduling that catches up within 30 minutes of the machine waking. Transcription runs on a self-built cross-platform CLI/GUI tool (Python, faster-whisper, FFmpeg, PyQt6) fully offline — no API cost or rate limits.

Admin & Product Management Platform
2026

Freelance, sole developer. Client and business logic covered by NDA; stack and design decisions only.

- **Transaction Integrity:** Root-caused a jOOQ / Spring integration defect — jOOQ acquired connections straight from the pool, outside Spring's transaction context, leaving `@Transactional` silently ineffective; rewired connection acquisition through `TransactionAwareDataSourceProxy` so multi-table writes roll back as a unit.
- **Caching Strategy:** Public query API uses conditional caching — query conditions as the cache key with filtering pushed down to SQL, and Caffeine capped by a byte-level weigher; admin edits and scheduled syncs evict the cache proactively, so data changes go live immediately without waiting on a TTL.
- **API Protection:** Four-tier rate limiting (global throttle → per-caller concurrency → daily quota → short-term rate), each tier with its own error code and a fixed check order so callers can self-diagnose rejections; global throttle parameters are externalized to configuration and tunable without redeploying.
- **Stack:** Spring Boot 2.7 (Java 11), jOOQ with code generation (client-mandated in place of JPA/Hibernate), MariaDB, Thymeleaf, AngularJS admin SPA, Docker Compose.

EDUCATION

BBA in International Business — National Dong Hwa University
– 2016

Java Web Development Program — Tibame
2022.08 – 2023.01